DevStack installation is given below:

Processor	RAM	Hard disk space
Intel/AMD processor with virtualisation support	4-8GB	50-100GB

To check the virtualisation support for the processor, use the following commands:

Processor family	Command
Intel	# grep -i "vmx" /proc/cpuinfo
AMD	# grep -i "svm" /proc/cpuinfo

Table 1: Hardware and OS flavours used (with minimum of 64GB RAM, 8 core CPU and 1TB hard disk)

Server model			
HP ProLiant DL585 G5			
Cisco-UCS- C-240-M4			
IBM System X3650 M4			
Dell Power Edge R730			
OS flavour	**Linux Kernel version**	**GNOME**	**LIBC**
Fedora 21 (64bit)	3.17.4	3.14	ldd (GNU libc) 2.21
CentOS 7 (64bit)	3.10.0-123.el7.x86_64	----	ldd (GNU libc) 2.17
Ubuntu 14.04 LTS (64bit)	2.19	----	ldd 2.19 EGLIBC

Note: Throughout this document 'Kilo/Liberty' releases of OpenStack/Devstack versions are used.

Troubleshooting

1. Whenever you create the *localrc* file as the root user and try to run the *stack.sh* command from the stack user, the error shown in Figure 5 will be encountered.
   Solution: When such an error is encountered, change the ownership of *devstack* directory to *stack* user, using the command given in Figure 6.
2. If you are using a proxy, while running the *stack.sh* command you may get the error shown in Figure 7.
   Solution: The best practice is to have an Internet connection without a proxy, which will help you avoid most of the issues.

```
#############################################################################
ENTER A PASSWORD TO USE FOR RABBIT.
#############################################################################
This value will be written to your localrc file so you don't have to enter it
again.  Use only alphanumeric characters.
If you leave this blank, a random default value will be used.
Enter a password now:
password
/stack.sh: line 594: /devstack/localrc: Permission denied
Error on exit
World dumping... see /opt/stack/logs/worlddump-2015-10-06-062308.txt for detail
df: '/run/user/1000/gvfs': Permission denied
/bin/sh: 1: brctl: not found
sudo: ebtables: command not found
stack@tempesttestingserver:/devstack$
```

Figure 5: *Localrc* permission denied

```
root@tempesttestingserver:/devstack# chown -R stack:stack /devstack/
root@tempesttestingserver:/devstack#
```

Figure 6: Ownership of stack user

```
ERROR] /devstack/functions-common:578 git call failed: [git clone git://git.openstack.org/openstack/requ
irements.git /opt/stack/requirements]
Error on exit
World dumping... see /opt/stack/logs/worlddump-2015-10-06-062824.txt for details
df: '/run/user/1000/gvfs': Permission denied
/bin/sh: 1: brctl: not found
sudo: ebtables: command not found
stack@tempesttestingserver:/devstack$
```

Figure 7: Proxy error

3. When we clone '*nova*' manually to */opt/stack/nova* from the Git repository and try to execute the command '*stack.sh*', the following exception will be encountered:

```
raise Exception ("Versioning for this project requires either
an sdist"
Exception: Versioning for this project requires either an
sdist tarball, or access to an upstream git repository. Are
you sure that git is installed?
```

   Solution: To avoid the above exception, copy the *.git* directory from the cloned *devstack* directory to the */opt/stack/nova* directory.
4. For downloading the earlier stable *DevStack* version from the Git repository, use the commands given below.
   Solution: Choose the stable version from the Git, for example, 'kilo' or any latest ('liberty') release.

```
$ git clone -b stable/kilo https://github.com/OpenStack-dev/
devstack.git
Cloning into 'devstack'...
remote: Counting objects: 28160, done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 28160 (delta 3), reused 0 (delta 0), pack-
reused 28146
Receiving objects: 100% (28160/28160), 9.59 MiB | 6.00 KiB/s,
done.
Resolving deltas: 100% (19543/19543), done.
```

DevStack installation on Ubuntu 14.04 LTS

After successfully installing Ubuntu 14.04 on the selected hardware, the following steps will configure and install DevStack and Tempest.

Note: Make sure that you have root user privileges, as most of the times, root user access is required.

Step 1: To get the latest version of Ubuntu packages, run the following command:

```
# apt-get update
```

Step 2: Install the Git package for downloading *devstack* files. Run the following command to install the Git package on the host machine:

```
#apt-get install git
```